Exercises from Ch. 1: Groups, Lang

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Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Exercise 0.1. (Exercise 1) Show that every group of order ≤ 5 is abelian.

Proof. First, we will take care of the cases where $|G|$ is prime (i.e. 2, 3, 5). We prove this by showing that the groups of these orders are cyclic and thus abelian (cyclic $\Rightarrow$ abelian). Suppose that $|G| = p \leq 5$, a prime. Then, by Lagrange's Theorem, for all $x \in G$, $|x| \bigm| |G|$. Hence, $|x| \bigm| p$. Therefore, $|x| = 1$ or $|x| = p$. If $|x| = 1$, then $x = e$ and we get $|G| = 1$ which is not a prime, but is also trivially abelian. If $|x| = p$, then let $G = \langle x \rangle$. Thus, $G = \left\{e = x^{|G|},\, x,\, x \cdot x = x^2\,, x \cdot x \cdot x = x^3,\, \dots\,, x^{|G|-1}\right\}$. Clearly, G is abelian since $x \cdot x^j = x^j \cdot x$ for all $x \in G$ and $1 \leq j \leq |G|$. Thus, if $|G| \leq 5$ is prime, then G is abelian. Next suppose $|G| = 4 = 2^2$. We prove a general case that all groups of order p^2 for a prime p are abelian and thus it would follow that G is abelian if $|G| = 4$.
Suppose $|G| = p^2$ for a prime p. Hence, G is a $p-group$ and hence $Z(G)$ is not trivial. Recall that the class equation is defined as

$$|G| = |Z(G)| + \sum_{x \in C} (G \colon G_x)$$

where C is a set of representatives for the distinct conjugacy classes and G_x is the isotropy group of x in G defined by $G_x = \{y \in G : \ x \cdot y = y\}$. But here we can consider G acting on itself by conjugation and thus $G_x = \{y \in G : \ yxy^{-1} = x\} = C_G(x) = \{g \in G : \ gx = xg\}$. Each term in $\sum\limits_{x \in C}(G \colon G_x) > 1$ and divides $|G| = p^2$. Hence, $\sum\limits_{x \in C}(G \colon G_x) \equiv 0 \pmod p$. Therefore, $|G| = |Z(G)| + kp$ ($k \in \mathbb{Z}^+$, and $kp = \sum (G \colon G_x)$). Then, $|Z(G)| = |G| - kp \Rightarrow |Z(G)| \equiv 0 \pmod p$. Thus, $p \bigm| |Z(G)| \Rightarrow |Z(G)| = p$ or p^2 (since $|Z(G)| \leq |G|$).
<u>Case 1:</u> Suppose $|Z(G)| = p^2$. Then $|Z(G)| = |G| \Rightarrow Z(G) = G$. Thus, G is abelian.
<u>Case 2:</u> Suppose $|Z(G)| = p$ Consider the quotient group $G/Z(G)$ (since $Z(G) \lhd G$). Thus, $|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{p^2}{p} = p$. Hence, $G/Z(G)$ is cyclic by above and thus $G/Z(G)$ is abelian. Now we will show that the fact that $G/Z(G)$ is abelian implies that G is abelian.
Since $G/Z(G)$ is cyclic, $G/Z(G) = \langle gZ(G) \rangle$. Thus, $xZ(G) = (gZ(G))^k = g^k Z(G)$ for some $k \in \mathbb{Z}$. By coset equality, $xZ(G) = g^k Z(G) \iff x(g^k)^{-1} \in Z(G) \iff x = g^k z_1$ for some $z_1 \in Z(G)$. Hence, every element of $x \in G$ can be written as $x = g^k z_1$ for some $k \in \mathbb{Z}$ and $z_1 \in Z(G)$. Similarly, let $y \in G$ and thus $y = g^\ell z_2$ for some $\ell \in \mathbb{Z}$ and $z_2 \in Z(G)$. We want to show that $xy = yx$. Then,

$$xy = g^k z_1 g^\ell z_2 = g^{k+\ell} z_1 z_2 = g^{\ell+k} z_2 z_1 = g^\ell z_2 g^k z_1 = yx$$

Therefore, G is abelian and note that we were able to commute elements since $z_1, z_2 \in Z(G)$. Thus, we have shown that if $G/Z(G)$ is cyclic, then G is abelian. We are done.

$\square$

Exercise 0.2. (Exercise 2)
Show that there are two non-isomorphic groups of order 4, namely the cyclic one, and the product of two cyclic groups of order 2.

Proof. Let G be a group of order 4. Let $g \in G$ such that $g \neq e$. By Lagrange's Theorem , $|g|\,\big|\,|G|$. Thus, $|g| = 2$ or 4. If $|g| = 4$, then $G = \langle g \rangle \cong \mathbb{Z}/4\mathbb{Z}$. If $|g| = 2$, then $G = \langle g \rangle \times \langle g \rangle \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. These are non-isomorphic since one is cyclic and one is not.

$\square$

Exercise 0.3. (Exercise 3)
Let G be a group. A **commutator** in G is an element of the form $aba^{-1}b^{-1}$ with $a, b \in G$. Let G^cbe the subgroup generated by the commutators. Then G^c is called the **commutator subgroup**. Show that G^c is normal. Show that any homomorphism of G into an abelian group factors through G/G^c.

Proof. By definition, $G^c = \{[a_1, b_1]^{c_1} \cdots [a_n, b_n]^{c_n} : n \in \mathbb{N},\ a_i, b_i \in G,\ c_i = \pm 1\}$. Thus, by construction, $G^c \leqslant G$. We show that this subgroup is normal. First, observe that for a single commutator,

$$\begin{aligned} g[a,b]g^{-1} &= g(aba^{-1}b^{-1})g^{-1} \\ &= ga(g^{-1}g)b(g^{-1}g)a^{-1}(g^{-1}g)b^{-1}g^{-1} \\ &= (gag^{-1})(gbg^{-1})(ga^{-1}g^{-1})(gb^{-1}g^{-1}) \\ &= \left[gag^{-1}, gbg^{-1}\right] \in G^c \end{aligned}$$

Conjugation is a homomorphism (distributes over products), and thus, since an element of G^c is a finite product of commutators, we have

$$\begin{aligned} g\left([a_1,b_1]^{c_1}[a_2,b_2]^{c_2}\cdots[a_n,b_n]^{c_n}\right)g^{-1} &= g\left(\left(a_1b_1a_1^{-1}b_1^{-1}\right)\cdots\left(a_nb_na_n^{-1}b_n^{-1}\right)\right)g^{-1} \\ &= g\left(a_1b_1a_1^{-1}b_1^{-1}\right)g^{-1}g(a_2b_2a_2^{-1}b_2^{-1})g^{-1}\cdots g(a_nb_na_n^{-1}b_n^{-1})g^{-1} \\ &= \left(g[a_1,b_1]g^{-1}\right)\left(g[a_2,b_2]g^{-1}\right)\cdots\left(g[a_n,b_n]g^{-1}\right) \end{aligned}$$

and our previous remarks, for all $1 \leq i \leq n$, $g[a_i, b_i]g^{-1} = \left[ga_ig^{-1}, gb_ig^{-1}\right]$. Then we get

$$g\left([a_1,b_1]\cdots[a_n,b_n]\right)g^{-1} = \left[ga_1g^{-1}, gb_1g^{-1}\right]\cdots\left[ga_ng^{-1}, gb_ng^{-1}\right] \in G^c$$

Note that since conjugation preserves products and inverses, we do not necessarily need to address the exponents $c_i = \pm 1$. Thus, $gG^cg^{-1} \subseteq G^c$ for all $g \in G$. Hence $G^c \lhd G$.
Let $f : G \to A$ be a homomorphism of G into an abelian group A. We want to show that the following diagram commutes:

$$\begin{tikzcd} G \arrow[rr, "f"] \arrow[dr, "\pi"'] & & A \\ & G/G^c \arrow[ur, "\bar{f}"'] & \end{tikzcd}$$

Here π is the canonical homomorphism (surjective). Observe that since f is a homomorphism and A is abelian,

$$\begin{aligned} f([a,b]) = f\left(aba^{-1}b^{-1}\right) &= f(a)f(b)f(a)^{-1}f(b)^{-1} \\ &= f(a)f(a)^{-1}f(b)f(b)^{-1} \\ &= e. \end{aligned}$$

Since every $x \in G^c$ is a finite product of commutators, by the fact that f is a homomorphism and by the above computation,

$$f(x) = e \quad \forall x \in G^c$$

Define

$$\bar{f} : G/G^c \to A$$

by $gG^c \mapsto f(g)$. First, we show that $\bar{f}$ is well-defined. Suppose $gG^c = hG^c$. Thus, by coset equality, $gh^{-1} \in G^c$. Then

$$
\begin{aligned}
f\left(gh^{-1}\right) &= f(g)f(h)^{-1} = e \\
&\Rightarrow f(g) = f(h)
\end{aligned}
$$

Therefore, $\bar{f}$ is well-defined. Now, let $gG^c, hG^c \in G/G^c$. Then

$$
\begin{aligned}
\bar{f}\left(\left(gG^c\right)\left(hG^c\right)\right) &= \bar{f}\left(ghG^c\right) \\
&= f(gh) \\
&= f(g)f(h) \\
&= \bar{f}\left(gG^c\right)\bar{f}\left(hG^c\right)
\end{aligned}
$$

Thus, $\bar{f}$ is a homomorphism. Hence, for all $g \in G$,

$$
\left(\bar{f} \circ \pi\right)(g) = \bar{f}(\pi(g)) = \bar{f}\left(gG^c\right) = f(g)
$$

Thus, $\bar{f} \circ \pi = f$. Therefore, every homomorphism of G into an abelian group factors through G/G^c. $\square$

Exercise 0.4. (Exercise 4)
Let H, K be subgroups of a finite group G with $K \subset N_H$. Show that

$$
|HK| = \frac{|H| \cdot |K|}{|H \cap K|}.
$$

Proof. Since $K \subseteq N_H$, we have $HK \leqslant G$. Define

$$
f\colon H \times K \to HK
$$

by $(h,k) = hk$. Let $x \in HK$. Then $x = hk$ for some $h \in H$ and $k \in K$. Hence, $f((h,k)) = hk = x$. Thus, f is surjective. Let $(h,k) \in \ker f$. Then

$$
\begin{aligned}
hk &= e \\
k &= h^{-1} \in K
\end{aligned}
$$

Thus, since $K \leqslant G$ and $h^{-1} \in K$, we must also have $h \in K$. Therefore

$$
\ker f = \left\{\left(h, h^{-1}\right) \in H \times K \colon h \in H \cap K\right\} \triangleleft H \times K
$$

We can define a bijection between $H \cap K$ and $\ker f$ by $h \mapsto \left(h, h^{-1}\right)$. Thus, $|\ker f| = |H \cap K|$. By the First Isomorphism Theorem,

$$
\begin{aligned}
H \times K/\ker f &\cong f\left(H \times K\right) \\
H \times K/H \cap K &\cong HK
\end{aligned}
$$

Thus,

$$
\begin{aligned}
|H \times K/H \cap K| &= |HK| \\
\frac{|H \times K|}{|H \cap K|} &= |HK| \\
\frac{|H| \cdot |K|}{|H \cap K|} &= |HK|
\end{aligned}
$$

$\square$

Exercise 0.5. (Exercise 6)
Prove that the group of inner automorphisms of a group G is normal in $\mathrm{Aut}(G)$.

Proof. Let $x \in G$. An inner automorphism is defined by $c_x(y) = xyx^{-1}$ for all $y \in G$. Something to note, perhaps irrelevant, is $f\colon G \to \mathrm{Aut}(G)$ by $x \mapsto c_x$ is a homomorphism onto $\mathrm{Inn}(G) \subseteq \mathrm{Aut}(G)$ with kernel $Z(G)$, so by the First Isomorphism Theorem,

$$G/Z(G) \cong \mathrm{Inn}(G) \subseteq \mathrm{Aut}(G).$$

Let $\varphi \in \mathrm{Aut}(G)$ and $c_x \in \mathrm{Inn}(G)$ for some $x \in G$. Let $y \in G$. Then

$$\begin{aligned}
\left(\varphi c_x \varphi^{-1}\right)(y) &= \varphi\left(c_x\left(\varphi^{-1}(y)\right)\right), \quad \varphi^{-1}(y) \in G \ \because \varphi \in \mathrm{Aut(G)} \\
&= \varphi\left(x\varphi^{-1}(y)x^{-1}\right) \\
&= \varphi(x)\varphi\left(\varphi^{-1}(y)\right)\varphi\left(x^{-1}\right) \\
&= \varphi(x)y\varphi(x)^{-1} \\
&= c_{\varphi(x)}(y) \in \mathrm{Inn}(G), \quad \because \varphi(x) \in G.
\end{aligned}$$

Hence, $\mathrm{Inn}(G) \triangleleft \mathrm{Aut}(G)$.

□

Exercise 0.6. (Exercise 7)
Let G be a group such that $Aut(G)$ is cyclic. Prove that G is abelian.

Proof. Suppose $\mathrm{Aut}(G)$ is cyclic. Hence, $\mathrm{Inn}(G)$ is cyclic since $\mathrm{Inn}(G) \subseteq \mathrm{Aut}(G)$. Consider the map $\psi : G \to \mathrm{Inn}(G)$ given by $g \mapsto \varphi_g$ for all $g \in G$ where $\varphi_g : G \to G$ by $x \mapsto gxg^{-1}$ for all $x \in G$. Notice that $\mathrm{Ker}\,\psi = Z(G)$, since if $\varphi_g(x) = gxg^{-1} = x$, then $gx = xg \Rightarrow g \in Z(G)$. Thus, by the First Isomorphism Theorem, $G/\mathrm{Ker}\,\psi \cong \mathrm{Inn}(G) \Rightarrow G/Z(G) \cong \mathrm{Inn}(G)$. Since we have this isomorphism, $G/Z(G)$ and $\mathrm{Inn}(G)$ have the same group structure, i.e. $G/Z(G)$ is cyclic and thus $G/Z(G)$ is abelian which implies from **Exercise 1** that G is abelian.

□

Exercise 0.7. (Exercise 9)

a) Let G be a group and H a subgroup of finite index. Show that there exists a normal subgroup N of G contained in H and also of finite index. [Hint: If $(G : H) = n$, find a homomorphism of G into S_n whose kernel is contained in H.]

b) Let G be a group and let H_1, H_2 be subgroups of finite index. Prove that $H_1 \cap H_2$ has finite index.

Proof. a) Let G be a group and $H \leqslant G$ such that $(G\colon H) = n$. Consider the left cosets $G/H = \{g_1H, \ldots, g_nH\}$. Define an action on these cosets by

$$\varphi_g\colon G/H \to G/H$$

by $\varphi_g(xH) = gxH$. This is well-defined for if $xH = yH$, then $x - y \in H$ and hence $gx - gy \in H \Rightarrow gxH = gyH \Rightarrow \varphi_g(xH) = \varphi_g(yH)$. The homomorphism φ_g can be shown to be bijective as well . Thus, $\varphi \in \mathrm{Sym}(G/H) \cong S_n$. Define

$$\psi\colon G \to S_n \cong \mathrm{Sym}(G/H)$$

by $\psi(g) = \varphi_g$. This is a homomorphism since for $g_1, g_2 \in G$,

$$\begin{aligned}
\psi(g_1g_2) &= \varphi_{g_1g_2}(xH) \quad \text{for any } xH \in G/H \\
&= g_1g_2xH \\
&= g_1\varphi_{g_2}(xH) \\
&= \varphi_{g_1}(\varphi_{g_2}(xH)) \\
&= \varphi_{g_1} \circ \varphi_{g_2}
\end{aligned}$$

The kernel, $\ker\psi$, is defined to be $g \in G$ such that $\psi(g) = id$. Hence, if $\psi(g) = id$ (i.e. $g \in \ker\psi$), we get

$$
\begin{aligned}
&\psi(g) = id \\
\Rightarrow &\varphi_g(xH) = xH \quad \forall xH \in G/H \\
\Rightarrow &gxH = xH \\
\Rightarrow &x^{-1}gxH = H \quad \forall x \in G \\
\Rightarrow &x^{-1}gx \in H \quad \forall x \in G \\
\Rightarrow &g \in xHx^{-1} \quad \forall x \in G \\
\Rightarrow &g \in \bigcap_{x\in G} xHx^{-1}
\end{aligned}
$$

Thus, $\ker\psi = \bigcap_{x\in G} xHx^{-1} \subseteq H$. Furthermore, $N = \ker\psi \lhd G$. By the First Isomorphism Theorem, $G/\ker\psi \cong \psi(G) \leqslant S_n$. Hence, $(G\colon N) = |G/N| = |\psi(G)| \leq |S_n| = n! < \infty$.

b) Suppose $H_1, H_2 \leqslant G$ such that $(G\colon H_1) = n$, $(G\colon H_2) = m$. Consider the cosets of $H_1 \cap H_2$. Any left coset of $H_1 \cap H_2$ has the form $g(H_1 \cap H_2) = gH_1 \cap gH_2$ for $g \in G$. Hence, every left coset of $H_1 \cap H_2$ is the intersection of a left coset of H_1 with a left coset of H_2. From $(G\colon H_1) = n$, $(G\colon H_2) = m$, there are at most nm distinct intersections and thus at most nm left cosets of $H_1 \cap H_2$. Hence, $(G\colon H_1 \cap H_2) \leq nm < \infty$.

□

Exercise 0.8. (Exercise 10) Let G be a group and let H be a subgroup of finite index. Prove that there is only a finite number of right cosets of H, and that the number of right cosets is equal to the number of left cosets.

Proof. By assumption, $(G\colon H) = n < \infty$. The group G acts on G/H in two ways:

$$\lambda : G \to \mathrm{Perm}(G/H)$$

by $x \mapsto x(gH)$ for some $g \in G$ and

$$\rho : G \to \mathrm{Perm}(G/H)$$

by $x \mapsto (gH)x$ for some $g \in G$. Consider $eH = H \in G/H$. The isotropy group of eH in G is

$$G_{eH} = \{g \in G :\ g(eH) = eH = H\} = H.$$

The orbit of eH under G is

$$G \cdot eH = \{g(eH) :\ g \in G\} = G/H.$$

By the orbit-stabilizer theorem and Lagrange's Theorem,

$$|G/H| = |G \cdot eH| = \underbrace{(G\colon G_{eH})}_{\text{under } \lambda} = (G\colon H) = n < \infty.$$

Now, under ρ, $G_{eH} = \{g \in G :\ (eH)g = H\}$. If $Hg = H$, then $g = eg \in Hg = H$, so $g \in H$. Conversely, if $g \in G$, then clearly $Hg \subseteq H$ and $g^{-1} \in H$ gives $H \subseteq Hg$, so $H = Hg$. Hence, $G_{eH} = H$. The orbit of eH under ρ in G is

$$G \cdot (eH) = \{(eH)g :\ g \in G\} = Hg = H/G \quad \text{(notation in Lang for right cosets)}$$

Again by the orbit-stabilizer theorem and Lagrange's

$$|H/G| = |G \cdot eH| = \underbrace{(G\colon G_{eH})}_{\text{under } \rho} = (G\colon H) = n < \infty.$$

Therefore, $|H/G| = |G/H| = n < \infty$, i.e. there is a finite number of right cosets and the number of left cosets is the same as the number of right cosets.

□

Exercise 0.9. (Exercise 11) Let G be a group and A a normal abelian subgroup. Show that G/A operates on A by conjugation, and in this manner get a homomorphism of G/A into $\mathrm{Aut}(A)$.

Proof. Define a map $\varphi : G/A \to \mathrm{Aut}(A)$ by $\varphi(gA) = gAg^{-1}$, $g \in G$. (Well-defined) Suppose that $gA = hA$. Then $h = ga \in H$ for some $a \in A$. For any $x \in A$, we have $hxh^{-1} = (ga)x(ga)^{-1} = g(axa^{-1})g^{-1}$. Since A is abelian, $axa^{-1} = x$. Thus, $hxh^{-1} = gxg^{-1}$. Thus, the definition of φ does not depend on the choice of representative in G/A. ($\mathrm{Im}\,\varphi \subseteq Aut(A)$) For each $g \in G$, the map $a \mapsto gag^{-1}$ is a homomorphism $A \to A$ since conjugation preserves products. It is bijective with inverse $a \mapsto g^{-1}ag$. Hence, $\varphi(gA) \in Aut(A)$. (Homomorphism) For $gA, hA \in G/A$ and $x \in A$,

$$\begin{aligned}(\varphi(gA) \circ \varphi(hA))(x) &= g(hxh^{-1})g^{-1} \\ &= (gh)x(gh)^{-1} = \varphi(ghA)(x).\end{aligned}$$

Therefore, φ is a group homomorphism. Thus, G/A acts on A by conjugation, and we obtain a homomorphism

$$\varphi : G/A \to \mathrm{Aut}(A)$$

Its kernel is $\mathrm{Ker}\,\varphi = C_G(A)/A$, where $C_G(A)$ is the centralizer of A in G.

□

Semidirect Product

Exercise 0.10. (Exercise 12) Let G be a group and let H, N be subgroups with N normal. Let γ_x be conjugation by an element $x \in G$.

a) Show that $x \mapsto \gamma_x$ induces a homomorphism $f : H \to Aut(N)$.

b) If $H \cap N = \{e\}$, show that the map $H \times N \to HN$ given by $(x, y) \mapsto xy$ is a bijection, and that this map is an isomorphism if and only if f is trivial, i.e. $f(x) = id_N$ for all $x \in H$.

c) We define G to be the <u>**semidirect product**</u> of H and N if $G = HN$ and $H \cap N = \{e\}$. Now, conversely, let N, H be groups, and let $\psi : H \to Aut(N)$ be a given homomorphism. Construct a semidirect product as follows. Let G be the set of pairs (x, h) with $x \in N$, $h \in H$. Define the composition law

$$(x_1, h_1)(x_2, h_2) = (x_1\psi(h_1)x_2, h_1h_2)$$

Show that this is a group law, and yields a semidirect product of N and H, identifying N with the set of elements $(x, 1)$ and H with the set of elements $(1, h)$.

Proof. a) First, let $e_H \in H$ be the identity element in H. Then, $f(e_H) = \gamma_{e_H}$. Let $n \in N$. Thus, $\gamma_{e_H}(n) = e_H n e_H^{-1} = n$ (since N normal). Thus, $f(e_H) = \gamma_{e_H} = id_N$. It is also well-defined since if $h_1 = h_2$, then $f(h_1) = \gamma_{h_1}(n) = h_1 n h_1^{-1}$ and $f(h_2) = \gamma_{h_2}(n) = h_2 n h_2^{-1}$. Clearly, we then have $f(h_1) = f(h_2)$. Let $x, y \in H$. Then, $f(xy) = \gamma_{xy}$. Take $n \in N$, $\gamma_{xy}(n) = xyn(xy)^{-1} = xyny^{-1}x^{-1} = x(\gamma_y(n))x^{-1} = \gamma_x(\gamma_y(n)) = (\gamma_x \circ \gamma_y)(n) = f(x) \circ f(y)$ as desired.

b) (Well-defined) For any $h \in H$, $n \in N$, $hn \in HN = \{xy : x \in H, y \in N\}$. (Injective) Suppose $(x_1, y_1), (x_2, y_2) \in H \times N$ and suppose $\varphi((x_1, y_1)) = \varphi((x_2, y_2)) \Rightarrow x_1y_1 = x_2y_2 \Rightarrow x_2^{-1}x_1 = y_2y_1^{-1}$. Note $x_2^{-1}x_1 \in H$ and $y_2y_1^{-1} \in N$. Thus, $x_2^{-1}x_1 \in H \cap N$ and $y_2y_1^{-1} \in H \cap N$. But $H \cap N = \{e\}$ by assumption, so $x_2^{-1}x_1 = e \Rightarrow x_1 = x_2$ and $y_2y_1^{-1} = e \Rightarrow y_1 = y_2$. Hence, $(x_1, y_1) = (x_2, y_2)$. (Surjective) Let $z \in HN$. Then, $z = hn$ for some $h \in H$, $n \in N$. Hence, $\varphi((h, n)) = hn = z$ with (h, n) clearly an element of $H \times N$. Thus, $\varphi : H \times N \to HN$ is a bijection.

Suppose $\varphi((x_1, y_1)(x_2, y_2)) = \varphi((x_1, y_1))\varphi((x_2, y_2))$ for $(x_i, y_i) \in H \times N$. This is if and only if

$$\begin{aligned} x_1x_2y_1y_2 &= x_1y_1x_2y_2 \\ \iff x_2y_1 &= y_1x_2 \quad \text{(left/right cancellation)} \\ \iff y_1 &= x_2y_1x_2^{-1} \end{aligned}$$

Since x_1, x_2, y_1, y_2 are arbitrary, this must hold for all $x_2 \in H$ and $y_1 \in N$. Hence, $\gamma_x \mid_N = \text{id}_N \Rightarrow f(x) = \text{id}_N$ for all $x \in H$.

c) Let $\psi : H \to Aut(N)$ be given by $\psi(h) = \varphi_h$ with $\varphi_h(n) = n' \in N$. Thus, we define the composition of $(x_1, h_1), (x_2, h_2) \in N \times H$ by

$$\begin{aligned} (x_1, h_1)(x_2, h_2) &= (x_1\psi(h_1)(x_2), h_1h_2) \\ &= (x_1\varphi_{h_1}(x_2), h_1h_2) \end{aligned}$$

(Identity) Let $e = (e_N, e_H) \in G = N \times H$. Then, $(x_1, h_1)(e_N, e_H) = (x_1\psi(h_1)(e_N), h \cdot e_N) = (x_1\varphi_{h_1}(e_N), h_1) = (x_1 \cdot e_N, h_1) = (x_1, h_1)$ (since $\varphi \in Aut(N)$ fixes the identity). Likewise, $(e_N, e_H)(x_1, h_1) = (x_1, h_1)$ and thus $e \in G$ which implies that $G \neq \varnothing$. (Closed) Observe that

$$\begin{aligned} (x_1, h_1)(x_2, h_2) &= (x_1\psi(h_1)(x_2), h_1h_2) \\ &= (x_1\varphi_{h_1}(x_2), h_1h_2) \end{aligned}$$

and since $\varphi_{h_1} \in Aut(N)$, $\varphi_{h_1}(x_2) \in N$. Hence, $x_1\varphi_{h_1}(x_2) \in N$ and $h_1h_2 \in H \Rightarrow (x_1\varphi_{h_1}(x_2), h_1h_2) \in G = N \times H$. (Associativity) Let $(x_1, h_1), (x_2, h_2), (x_3, h_3) \in G$. Then,

$$\begin{aligned} (x_1, h_1)\left[(x_2, h_2)(x_3, h_3)\right] &= (x_1, h_1)(x_2\psi(h_2)(x_3), h_2h_3) \\ &= (x_1, h_1)(x_2\varphi_{h_2}(x_3), h_2h_3) = (x_1\psi(h_1)\left[x_2\varphi_{h_2}(x_3)\right], h_1h_2h_3) \\ &= (x_1\varphi_{h_1}(x_2\varphi_{h_2}(x_3)), h_1h_2h_3) \\ &= (x_1\varphi_{h_1}(x_2)\varphi_{h_1}(\varphi_{h_2}(x_3)), h_1h_2h_3) \end{aligned}$$

Now,

$$\begin{aligned} \left[(x_1, h_1)(x_2, h_2)\right](x_3, h_3) &= (x_1\psi(h_1)(x_2), h_1h_2)(x_3, h_3) \\ &= (x_1\varphi_{h_1}(x_2), h_1h_2)(x_3, h_3) \\ &= (x_1\varphi_{h_1}(x_2)\psi(h_1h_2)(x_3), h_1h_2h_3) \\ &= (x_1\varphi_{h_1}(x_2)(\psi(h_1)\psi(h_2))(x_3), h_1h_2h_3) \\ &= (x_1\varphi_{h_1}(x_2)\varphi_{h_1}(\varphi_{h_2}(x_3)), h_1h_2h_3) \end{aligned}$$

Thus, $(x_1, h_1)\left[(x_2, h_2)(x_3, h_3)\right] = \left[(x_1, h_1)(x_2, h_2)\right](x_3, h_3)$. (Inverses) Suppose that $(x_1, h_1)(x_2, h_2) = (e_N, e_H)$. Thus,

$$\begin{aligned} (x_1, h_1)(x_2, h_2) &= (e_N, e_H) \\ \Rightarrow (x_1\psi(h_1)(x_2), h_1h_2) &= (e_N, e_H) \\ \Rightarrow (x_1\varphi_{h_1}(x_2), h_1h_2) &= (e_N, e_H) \\ \Rightarrow x_1\varphi_{h_1}(x_2) = e_N &\quad \text{and} \quad h_1h_2 = e_H \\ \Rightarrow \varphi_{h_1}(x_2) = x_1^{-1} &\qquad \Rightarrow h_2 = h_1^{-1} \in H \\ \because \varphi_{h_1} \in Aut(N), \varphi_{h_1^{-1}} &\in Aut(N) \\ \Rightarrow \varphi_{h_1^{-1}}(\varphi_{h_1}(x_2)) = \varphi_{h_1^{-1}}(x_1^{-1}) &\Rightarrow x_2 = \varphi_{h_1^{-1}}(x_1^{-1}) \in N. \end{aligned}$$

Therefore, $(x_1, h_1)^{-1} = (\varphi_{h_1^{-1}}(x_1^{-1}, h_1^{-1}) = (\psi(h_1^{-1})(x_1^{-1}), h_1^{-1}) \in G$. $(G = HN)$ Let $(x, h) \in G = N \times H$. Then,

$$\begin{aligned}(x, h) &= (x\varphi_{e_H}(e_N), e_H \cdot h)\\ &= (x\psi(e_H)(e_N), e_H \cdot h)\\ &= \underbrace{(x, e_H)}_{\in N}\underbrace{(e_N, h)}_{\in H}\end{aligned}$$

Thus, $G = NH$. $(N \cap H = \{e\})$ Note that $(x, e_H) \in H \iff x \in e_N$ and $(e_N, h) \in N \iff h = e_H$. Thus, $N \cap H = \{(e_N, e_H)\} = \{e\}$. Therefore,

$$G = N \rtimes_\psi H.$$

□

Exercise 0.11. (Exercise 13)

a) Let H, N be normal subgroups of a finite group G. Assume that the orders of H, N are relatively prime. Prove that $xy = yx$ for all $x \in H$ and $y \in N$, and that $H \times N \cong HN$.

b) Let $H_1, \ldots, H_r$ be normal subgroups of G such that the order of H_i is relatively prime to the order of H_j for $i \neq j$. Prove that

$$H_1 \times \cdots \times H_r \cong H_1 \cdots H_r.$$

Example. If the Sylow subgroups of a finite group are normal, then G is the direct product of its Sylow subgroups.

Proof. a) Assume $H, N \lhd G$ with $(|H|, |N|) = 1$. Consider the commutator,

$$[x, y] := xyx^{-1}y^{-1} \quad \forall x \in H, y \in N$$

Thus, $xyx^{-1}y^{-1} \in H$ since $yx^{-1}y^{-1} \in H$ by the fact that $H \lhd G$ and therefore $x(yx^{-1}y^{-1}) \in H$. Similarly $xyx^{-1}y^{-1} \in N$ since $xyx^{-1} \in N$ by the fact that $N \lhd G$. Thus, $[x, y] = xyx^{-1}y^{-1} \in H \cap N$. By Lagrange's Theorem,

$$|H \cap N| \Big| |H| \quad \text{and} \quad |H \cap N| \Big| |N|$$

and thus,

$$|H \cap N| \Big| \underbrace{(|H|, |N|)}_{=1}$$

Hence, we must have $|H \cap N| = 1 \Rightarrow H \cap N = \{e\}$. Then since $[x, y] \in H \cap N$, it follows that $xyx^{-1}y^{-1} = e \Rightarrow xy = yx$. Finally, by *Exercise 12*, $H \times N \cong HN$.

b) Assume $H_1, \ldots, H_r \lhd G$ with $(|H_i|, |H_j|) = 1$ if $i \neq j$. From *part (a)*, for any H_i, H_j with $i \neq j$, $H_i \cap H_j = \{e\}$ and $xy = yx$ for all $x \in H_i$, $y \in H_j$. Hence, $H_i \times H_j \cong H_iH_j$. We proceed by induction. Suppose that $H_1 \times \cdots \times H_{r-1} \cong H_1 \cdots H_{r-1}$. Then, $(|H_r|, |H_1 \cdots H_{r-1}|) = 1$ and $H_r \lhd G$. Thus, by the same argument as above, $H_1 \times \cdots \times H_r \cong H_1 \cdots H_r$.

□

Exercise 0.12. (Exercise 14)
Let G be a finite group and let N be a normal subgroup such that N and G/N have relatively prime orders.

a) Let H be a subgroup of G having the same order as G/N. Prove that $G = HN$.

b) Let g be an automorphism of G. Prove that $g(N) = N$.

Proof. Let G be a finite group of order n and let $N \lhd G$ such that $(|N|, |G/N|) = 1$.

a) Assume that $H \leqslant G$ such that $|H| = |G/N| = (G\colon N)$. Then $(|H|, |N|) = 1$. We have

$$
\begin{aligned}
|HN| &= \frac{|H| \cdot |N|}{|H \cap N|} \\
&= \frac{|G/N| \cdot |N|}{|H \cap N|} \\
&= \frac{\frac{|G|}{|N|} \cdot |N|}{|H \cap N|} \\
&= \frac{|G|}{|H \cap N|}
\end{aligned}
$$

Then, if $|HN| = |G|$, we would require that $|H \cap N| = 1 \Rightarrow H \cap N = \{e\}$. Hence, since $HN \subseteq G$, we have that $G = HN$ if and only if $H \cap N = \{e\}$. Thus, we show the converse of this biconditional.

By Lagrange's Theorem, $|H \cap N| \Big| |H|$ and $|H \cap N| \Big| |N|$, and thus $|H \cap N| \Big| \underbrace{(|H|, |N|)}_{=1}$. Hence, $|H \cap N| = 1$ and therefore, $H \cap N = \{e\}$. It follows from the above biconditional that $G = HN$.

b) Let $g \in \mathrm{Aut}(G)$. Then $g(N) \leqslant G$ with $|g(N)| = |N|$. By our assumption, $(|N|, |G/N|) = 1$, we have $(|g(N)|, |G/g(N)|) = 1$. Let $H \leqslant G$ such that $|H| = |G/N|$ as in *part (a)* which implies that $G = HN$ and $H \cap N = \{e\}$. Then $|g(H)| = |H| = |G/N|$ and applying *part (a)* to the normal subgroup $g(N)$ gives that $G = g(H) \cdot g(N)$ and $g(H) \cap g(N) = \{e\}$. Hence, since $N, g(N) \lhd G$ are the same order, admitting a complement of order $|G/N|$. The order of $Ng(N)$ is

$$
|Ng(N)| = \frac{|N| \cdot |g(N)|}{|N \cap g(N)|} = \frac{|N|^2}{|N \cap g(N)|} \leq |G| = |N| \cdot |G/N|
$$

Hence, we have that $|N \cap g(N)| \geq \frac{|N|}{|G/N|}$. Since $(|N|, |G/N|) = 1$, this implies that $|N \cap g(N)| = |N|$, so $N \subseteq g(N)$. By symmetry, using g^{-1} (since g is an automorphism), we have $g(N) \subseteq N$. Therefore, $g(N) = N$.

□

Some operations

Exercise 0.13. (Exercise 16)
Let H be a proper subgroup of a finite group G. Show that G is not the union of all the conjugates of H. (But see Exercise 23 of Chapter XIII.)

Proof. Let $H \leqslant G$ be proper, that is, $H \neq G$. Hence, $|H| < |G|$. By way of contradiction, assume that $G = \bigcup_{g \in G} gHg^{-1}$. Suppose that there are n distinct conjugates H_i, so $G = \bigcup_{i=1}^{n} H_i$. Then

$$
|G| = \left| \bigcup_{i=1}^{n} H_i \right|.
$$

However, since each conjugate contains e, the conjugates are not disjoint. Hence,

$$
|G| < \left| \bigcup_{i=1}^{n} H_i \right| = \sum_{i=1}^{n} H_i = n \cdot |H|.
$$

Furthermore, $n = (G\colon N_G(H))$ and thus

$$\begin{aligned}|G| &< n\cdot|H|\\ |G| &< (G\colon N_G(H))\cdot|H|\\ |G| &< \frac{|G|}{|N_G(H)|}\cdot|H|\\ |N_G(H)| &< |H|\end{aligned}$$

which is a contradiction since $H \subseteq N_G(H)$. Therefore,

$$G \neq \bigcup_{g\in G} gHg^{-1}.$$

□

Throughout, p is a prime number.

Exercise 0.14. (Exercise 20) Let P be a p-group. Let A be a normal subgroup of order p. Prove that A is contained in the center of P.

Proof. Assume $|A| = p$. Thus, $A \cong C_p$, the cyclic group of order p (since prime order implies cyclic). Thus, $\mathrm{Aut}(A)$ is cyclic of order $p-1$. Define

$$\varphi : P \to \mathrm{Aut}(A)$$

by $g \mapsto c_g$, conjugation by g in A. Since $A \triangleleft P$, and by previous exercises, c_g is a bijective homomorphism of A into itself, hence $c_g \in \mathrm{Aut}(A)$. Our map φ is a homomorphism since for $g, h \in P$ we have

$$\varphi(gh) = c_{gh} = c_g \circ c_h = \varphi(g)\varphi(h).$$

By definition, $\varphi(P) \leq \mathrm{Aut}(A)$, thus $|\varphi(P)|\Big||\mathrm{Aut}(A)| \Rightarrow |\varphi(P)|\Big|p-1$. But $\varphi(P)$ is also a homomorphic image of the p-group P, hence $|\varphi(P)| = p^k$ for some k. This is due to the First Isomorphism Theorem which states

$$P/\ker\varphi \cong \varphi(P).$$

Since $|P| = p^n$ for some n by assumption, and $\ker\varphi \leq P$, by Lagrange's Theorem, $|\ker\varphi| = p^k$ for $k \leq n$. Then, by counting sizes via the isomorphism,

$$|\varphi(P)| = |P/\ker\varphi| = \frac{|P|}{|\ker\varphi|} = \frac{p^n}{p^k} = p^{n-k} \qquad \text{(a power of } p\text{)}$$

Whence, the only positive integer that is both a power of p and a divisor $p-1$ is 1. Thus, $|\varphi(P)| = p^0 = 1$. Thus means $\varphi(P)$ is trivial, i.e. $\varphi(g) = \mathrm{id}_A$ for all $g \in P$. Then, $(\varphi(g))(a) = c_g(a) = gag^{-1} = a$ (since $\varphi(g)(a) = a$ by $\varphi(g) = \mathrm{id}_A$). Hence, $ga = ag$. Therefore every element of A commutes with every element of P. Therefore,

$$A \subseteq Z(P).$$

□

Exercise 0.15. (Exercise 21) Let G be a finite group and H a subgroup. Let P_H be a p-Sylow subgroup of H. Prove that there exists a p-Sylow subgroup P of G such that $P_H = P \cap H$.

Proof. Let G be a finite group, $H \leq G$, and let P_H be a p-Sylow subgroup of H. By Theorem 6.4, there exists a p-Sylow subgroup $P \leq G$ such that every p-subgroup of G is contained in P. Hence, since $H \leq G$, P_H is a p-subgroup of G (p-Sylow subgroup of H, by assumption) and thus

$$P_H \subseteq P$$

Furthermore, $P_H \subseteq H \Rightarrow P_H \subseteq P \cap H$. Observe that $P \cap H \leq H$ and thus since P is a p-subgroup (also Sylow), $P \cap H$ is a p-subgroup of H and therefore $P \cap H \subseteq P_H$. Hence,

$$P \cap H = P_H.$$

$\square$

Exercise 0.16. (Exercise 22)
Let H be a normal subgroup of a finite group G and assume that $\#(H) = p$. Prove that H is contained in every p-Sylow subgroup of G.

Proof. Assume that $H \triangleleft G$ and suppose $|H| = p$. This follows nicely from all parts of Theorem 6.4. Thus, H is a p-subgroup and for all $x \in G$,

$$xHx^{-1} = H \quad \text{(normality)}$$

Then by definition, $N(H) = G$. By Theorem 6.4(i), since H is a p-subgroup of G, H is contained in some p-Sylow subgroup, say K. Then, by Theorem 6.4(ii), all p-Sylow subgroups are conjugate and thus $K' = gKg^{-1}$ is a p-Sylow subgroup for some $g \in G$. Conjugating the inclusion $H \subseteq K$, we get

$$H = xHx^{-1} \subseteq xKx^{-1} = K'$$

Hence, $H \subseteq K'$ with K' a different, unless K normal, p-Sylow subgroup. Thus, we can continue this process until we get all $pk+1$ p-Sylow subgroups ($k \in \mathbb{Z}$, by Theorem 6.4(iii) which states the number of p-Sylow subgroups of G is $\equiv 1 \mod p$). Thus, it follows that H is contained in every p-Sylow subgroup.

$\square$

Exercise 0.17. (Exercise 23) Let P, P' be p-Sylow subgroups of a finite group G.

a) If $P' \subseteq N(P)$ (normalizer of P), then $P' = P$.

b) If $N(P') = N(P)$, then $P' = P$.

c) We have $N(N(P)) = N(P)$.

Proof. a) Suppose $P' \subseteq N(P)$. Then for all $x \in P'$, we have

$$xPx^{-1} = P$$

and thus $P \triangleleft P'$. Since P, P' are p-groups and P is a normal subgroup of P', the quotient P'/P is also a p-group. By definition, $|P| = |P'| = p^n$ with p^n the highest power of p dividing $|G|$. Hence, $|P'/P| = \frac{|P'|}{|P|} = 1$. Thus, $P = P'$.

b) We know $P' \subseteq N(P')$ and $P \subseteq N(P)$. By assumption $N(P') = N(P)$, so

$$P' \subseteq N(P') = N(P)$$

Thus, by part (a), $P' = P$.

c) ($N(P) \subseteq N(N(P))$) This inclusion follows trivially from the fact that for any subgroup $H \leqslant G$, $H \subseteq N(H)$, i.e. every subgroup is a subset of its normalizer. Thus, if you let $K = N(H)$, then we have $K \subseteq N(K)$.

($N(N(P)) \subseteq N(P)$) Let $x \in N(N(P))$. Then

$$xN(P)x^{-1} = N(P)$$

and in particular,

$$xPx^{-1} \subseteq N(P)$$

Thus, xPx^{-1} is a p-Sylow subgroup of $N(P)$. Since $P \subseteq N(P)$, P is also a p-Sylow subgroup of $N(P)$. By Theorem 6.4, there exists $y \in N(P)$ such that

$$y(xPx^{-1})y^{-1} = P$$

Then $yx \in N(P)$ and since $N(P)$ is a subgroup, we have that $x \in N(P)$ as was to be shown. Therefore, $N(N(P)) = N(P)$.

□

Explicit determination of groups

Exercise 0.18. (Exercise 24)
Let p be a prime number. Show that a group of order p^2 is abelian, and that there are only two such groups up to isomorphism.

Proof. Suppose $|G| = p^2$ for a prime p. Thus, G is a p-group and hence $Z(G)$ is not trivial. Recall that the class equation is defined as

$$|G| = |Z(G)| + \sum_{x \in C} (G\colon G_x)$$

where C is a set of representatives for the distinct conjugacy classes and G_x is the isotropy group of x in G defined as $G_x = \{x \in G : \ xy = y\}$ with $y \in G$ (here G is acting on itself via conjugation). Each term in $\sum_{x \in C} (G\colon G_x) > 1$ and divides $|G| = p^2$. Hence, $\sum_{x \in C} (G\colon G_x) \equiv 0 \pmod p$. Thus,
$|G| = |Z(G)| + kp$ $(k \in \mathbb{Z}^+)$ $\left[kp = \sum_{x \in C} (G\colon G_x)\right]$. Therefore, $|Z(G)| = |G| - kp \Rightarrow |Z(G)| \equiv 0 \pmod p$.
Thus, $p \,\big|\, |Z(G)| \Rightarrow |Z(G)| = p$ or p^2.
But by **Exercise 1**, Cases 1 and 2 cover this, thus, it follows that G is abelian.
If G has an element g of order p^2, then the order of $\langle g \rangle$ is p^2 and thus $G = \langle g \rangle$. Therefore, in this case, $G \cong \mathbb{Z}/p^2\mathbb{Z}$.
If no element (not including the identity) of G has order p^2, then it must have order p. Choose $a \in G$ such that $|\langle a \rangle| = p$ and thus $(G\colon \langle a \rangle) = p$. Choose $b \notin \langle a \rangle$. Hence, we also have $|b| = p$. Hence, we have $\langle a \rangle \cap \langle b \rangle = \{e\}$ with the product $\langle a \rangle \times \langle b \rangle$ having order p^2. Therefore, $G = \langle a \rangle \times \langle b \rangle \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$.
Then these two possibilities of G can not be isomorphic since one is cyclic and one is not, and cyclicity is invariant under isomorphisms.

□

Exercise 0.19. (Exercise 25)
Let G be a group of order p^3, where p is prime, and G is not abelian. Let Z be its center. Let C be a cyclic group of order p.

a) Show that $Z \cong C$ and $G/Z \cong C \times C$.

b) Every subgroup G of order p^2 contains Z and is normal.

c) Suppose $x^p = 1$ for all $x \in G$. Show that G contains a normal subgroup $H \cong C \times C$.

Proof. a) Assume that G is not an abelian group with order p^3. Since $|G| = p^3$, G is a p-group. By Theorem 6.5, since $p^3 > 1$, G has a nontrivial center. By the assumption that G is not abelian, $G \neq Z$. Thus, $|Z| = p$ or p^2. If $|Z| = p^2$, then $|G/Z| = p$ and thus G/Z is cyclic. By **Exercise 1**, it follows that G is abelian, a contradiction! Therefore $|Z| = p$ and thus Z is cyclic of order p, i.e. $Z \cong C$. Then $|G/Z| = p^2$. By **Exercise 24**, there are two abelian groups, up to isomorphism, of order p^2, a cyclic group and a non-cyclic group. If G/Z is cyclic, then by **Exercise 1**, G is abelian, a contradiction! Hence, G/Z is not cyclic which implies that $G/Z \cong C \times C$.

b) Let $H \leqslant G$ such that $|H| = p^2$. Thus, by **Exercise 24**, H is abelian. It follows that $H \cong C_{p^2} \cong \mathbb{Z}/p^2\mathbb{Z}$ or $H \cong C \times C$. From part (a), $|Z| = p$. Hence, $|G/Z| = p^2$. In part (a), we showed that

$$G/Z \cong C \times C.$$

Consider

$$\pi\colon G \to G/Z$$

given by $g \mapsto gZ$. Thus

$$\ker \pi = \{x \in G\colon xZ = Z\} = G \cap Z = Z$$

Let $\pi|_H$ be the restriction to H. Then

$$\ker \pi|_H = H \cap Z \subseteq Z.$$

Since $\ker \pi|_H \lhd H$, Lagrange's and the containment, $|H \cap Z|\big||Z| \Rightarrow |H \cap Z| = 1$ or p. By the First Isomorphism Theorem,

$$\begin{aligned} &H/\ker \pi|_H \cong \pi(H) \\ \Rightarrow &H/H \cap Z \cong \pi(H) \end{aligned}$$

Therefore,

$$|H/H \cap Z| = |\pi(H)| \Rightarrow \frac{|H|}{|H \cap Z|} = |\pi(H)|.$$

If $|H \cap Z| = 1$, then $|\pi(H)| = |H| = p^2 = |G/Z|$. This would mean that H maps surjectively onto G/Z but $H \leqslant G$. Thus, we must have $|H \cap Z| = p$. Hence since $Z \cong C$, we have $H \cap Z = Z$ and therefore $Z \subseteq H$. Then $|\pi(H)| = p$. Thus, $\pi(H) \leqslant G/Z \cong C \times C$. Since G/Z is abelian , $\pi(H) \lhd G/Z$. The pre-image of a normal subgroup is normal, hence $\pi^{-1}(\pi(H)) \lhd G$. The pre-image is defined as

$$\begin{aligned} \pi^{-1}(\pi(G)) &= \{x \in G\colon xZ \in \pi(H)\} \\ &= \{x \in G\colon xZ = yZ \quad \text{for some } h \in H\} \end{aligned}$$

Let $g \in \pi^{-1}(\pi(H))$. Thus, $\pi(g) \in \pi(H)$ which implies that $\pi(g) = \pi(h)$ for some $h \in H$. Then $gZ = hZ$. Hence, there exists some $x \in Z$ such that $g = hx$. Since $Z \subseteq H$, we have $g = hx \in H$. Hence, $\pi^{-1}(\pi(H)) \subseteq H$. Now let $h \in H$. Then $\pi(h) = hZ \in \pi(H)$. Thus, $H \subseteq \pi^{-1}(\pi(H))$. Therefore, $\pi^{-1}(\pi(H)) = H$. Then from

$$\pi^{-1}(\pi(H)) \lhd G,$$

we have $H \lhd G$. Therefore, $Z \subseteq H$ and H is normal in G.

c) Since $G/Z \cong C \times C$ from part (a), G/Z contains a cyclic subgroup of order p, say

$$\langle aZ \rangle \leqslant G/Z.$$

Let

$$\begin{aligned} H = \pi^{-1}\left(\langle aZ \rangle\right) &= \{x \in G\colon \pi(x) \in \langle aZ \rangle\} \\ &= \{x \in G\colon xZ = a^k Z,\ k \in \mathbb{Z}\} \\ &= \{x \in G\colon x = a^k z,\ k \in \mathbb{Z},\ z \in Z\}. \end{aligned}$$

Thus, $|H| = |\langle aZ \rangle| \cdot |Z| = p^2$. By part (b), $H \lhd G$ and $Z \subseteq H$. By assumption, $x^p = 1$ for all $x \in G$, so all elements of H have orders dividing p. Therefore, since $|H| = p^2$ and no element of H has order p^2 ($p^2 \nmid p$), H cannot be cyclic. Thus, by **Exercise 24**, $H \cong C \times C$.

$\square$

Exercise 0.20. (Exercise 26)

a) Let G be a group of order pq, where p, q are primes and $p < q$. Assume that $q \not\equiv 1 \mod p$. Prove that G is cyclic.

b) Show that every group of order 15 is cyclic.

Proof. a) Let $|G| = pq$ with $p < q$ distinct primes and assume that $q \not\equiv 1 \mod p$. Thus, there exists p, q-Sylow subgroups H_p, H_q respectively with $|H_p| = p$ and $|H_q| = q$. By Lemma 6.7, since $(G \colon H_q) = p < q$ is the smallest prime dividing the order of G, we have $H_q \triangleleft G$. Furthermore, since H_p, H_q are both prime order, they are cyclic (Proposition 4.1) and thus there exists $x \in H_p$ and $y \in H_q$ such that $H_p = \langle x \rangle$ and $H_= \langle y \rangle$. Define

$$\varphi \colon H_p \to \operatorname{Aut}(H_q)$$

by $x \mapsto \varphi_x$ with $\varphi_x \colon H_q \to H_q$ being conjugation by x, an automorphism. Since H_q is cyclic, any automorphism must send a generator to another generator, i.e. $y \mapsto y^k$. Thus, $k \in (\mathbb{Z}/q\mathbb{Z})^\times$. Then

$$\operatorname{Aut}(H_q) \cong (\mathbb{Z}/q\mathbb{Z})^\times$$

which implies that $|\operatorname{Aut}(H_q)| = q - 1$. Then

$$\begin{aligned} \ker \varphi &= \{x \in H_p \colon \varphi_x(y) = y, \ \forall\, y \in H_q\} \\ &= \{x \in H_p \colon xy = yx, \ \forall\, y \in H_q\} \triangleleft H_p \end{aligned}$$

By Lagrange's Theorem, $|\ker \varphi| \Big| |H_p|$. Hence $|\ker \varphi| = 1$ or p. If $|\ker \varphi| = 1$, by the First Isomorphism Theorem, we have

$$H_p / \ker \varphi \cong \varphi(H_p) \leqslant \operatorname{Aut}(H_q)$$

and thus $|H_p / \ker \varphi| = |H_p| = p = |\varphi(H_p)|$. Then, by Lagrange's Theorem, $|\varphi(H_p)| \Big| |\operatorname{Aut}(H_q)|$, but by assumption, $q \not\equiv 1 \mod p$. Hence we can not have $|\ker \varphi| = 1$. Thus, $|\ker \varphi| = p$ and therefore by the First Isomorphism Theorem, $|H_p / \ker \varphi| = 1 = |\varphi(H_p)|$. Thus, $\varphi(H_p) = \{\text{id.}\}$. Since φ is trivial, every element of H_p commutes with every element of H_q. Thus, $H_p H_q$ is abelian. Let $z \in H_p \cap H_q$. Then $z \in H_p$ and $z \in H_q$. Hence, $z = x^k$ and $z = y^l$ for some $k \in (\mathbb{Z}/p\mathbb{Z})^\times$ and $l \in (\mathbb{Z}/q\mathbb{Z})^\times$. Then

$$\begin{aligned} x^k &= y^l \\ (x^k)^p &= (y^l)^p \\ e &= y^{lp} \end{aligned}$$

Thus, $q \mid lp$. By Euclid's Lemma, since q is a prime and $q \nmid p$, we have $q \mid l$ which implies that $l \equiv 0 \mod q$ and $y^l = e$. Then $x^k = y^l = e \Rightarrow k \equiv 0 \mod p$ and therefore $z = e$. Hence $H_p \cap H_q = \{e\}$. Thus,

$$|H_p H_q| = \frac{|H_p| \cdot |H_q|}{|H_p \cap H_q|} = \frac{pq}{1} = pq = |G|$$

Then $G = H_p H_q$. Since H_p, H_q are cyclic with $(p, q) = 1$, H_p, H_q abelian, and $H_p \cap H_q = \{e\}$, we have $H_p H_q = \langle xy \rangle$ with $|xy| = \operatorname{lcm}(|x|, |y|) = \operatorname{lcm}(p, q) = pq = |G|$. Therefore, $\langle xy \rangle$ has order pq and thus $G = H_p H_q = \langle xy \rangle$ is cyclic.

b) Let $|G| = 15 = 3 \cdot 5$. Since $5 \not\equiv 1 \mod 3$, by part (a), G is cyclic.

□

Exercise 0.21. (Exercise 28) Let p, q be distinct primes. Prove that a group of order p^2q is solvable, and that one of its Sylow subgroups is normal.

Proof. Let n_p and n_q denote the number of p-Sylow and q-Sylow subgroups G, respectively. By Sylow's Theorems,

$$n_p \Big| q, \quad n_p \equiv 1 \pmod p, \quad n_q \Big| p^2, \quad n_q \equiv 1 \pmod q$$

Hence,

$$n_p \in \{1, q\}, \quad n_q \in \{1, p, p^2\}$$

Case 1: $(p < q)$ If $n_q = p$, then $q \Big| p - 1$ which is impossible since $q > p - 1$. If $n_q = p^2$, then $q \Big| (p-1)(p+1)$ which is impossible since $q > p + 1$ (as well). Thus, $n_q = 1$, so the q-Sylow subgroup $Q \triangleleft G$ is normal. Case 2: $(p > q)$ If $n_p = q$, then $p \Big| q - 1$ which is impossible since $p > q - 1$. Thus, $n_p = 1$, so the p-Sylow subgroups $P \triangleleft G$ is normal. In both cases, G has a normal Sylow subgroup N (either $|N| = q$ or $|N| = p^2$). Now consider the quotient G/N.

- $|N|$ is a prime power and therefore is solvable.
- $|G/N|$ is either p^2 or pq and in either case, both are known to be solvable.

Since both N and G/N are solvable, it follows that G is solvable. [The fact that $n_p, n_q = 1 \Rightarrow P, Q$ are normal is since all Sylow groups are conjugate and if there is only one Sylow subgroup, then it is fixed by conjugation by G, hence normal.]

□

Exercise 0.22. (Exercise 29)
Let p, q be odd primes. Prove that a group of order $2pq$ is solvable.

Proof. Let G be a group of order $2pq$ and without loss of generality suppose that $q < p$. Thus, we have $2 < q < p$. We need a lemma.

Lemma 0.1. If p, q, r are any three prime numbers, then a group of order pqr is not simple.

Proof. Recall that a group is said to be *simple* if the group has no nontrivial normal subgroups (i.e. the trivial subgroup and the group itself are the only normal subgroups).
Case 1: $p = q = r$: If $p = q = r$, then $|G| = p^3$. By **Exercise 25**, we see at once that G contains a nontrivial normal subgroup and thus is not simple.
Case 2: $p = q \neq r$: If $p = q \neq r$, then $|G| = p^2r$. Let n_p denote the number of p-Sylow subgroups of G and n_r denote the number of r-Sylow subgroups of G. Thus, by Theorem 6.4,

$$n_p \mid r, \ n_p \equiv 1 \mod p \quad \text{and} \quad n_r \mid p^2, \ n_r \equiv 1 \mod r$$

Thus, $n_p \in \{1, r\}$. If $n_p = 1$, let P denote the unique p-Sylow subgroup of G. Observe that since G is not a p-group, $P \neq G$. Then since p-Sylow subgroups are conjugate and $n_p = 1$, it follows that for any $g \in G$, $gPg^{-1} = P$. Hence, $P \triangleleft G$. Therefore G is not simple. Hence, assume that $n_p = r$. Thus, $r \equiv 1 \mod p$, so $r > p$. Since $n_r \mid p^2$ and $n_r \equiv 1 \mod r$, $n_r \in \{1, p, p^2\}$. If $n_r = 1$, then by the same logic as in the case for $n_p = 1$, G has a normal r-Sylow subgroup and therefore G is not simple. Therefore, suppose that $n_r \neq 1$. If $n_r = p$, then $p \equiv 1 \mod r$ which implies that $p > r$, a contradiction! Thus, we must have $n_r = p^2$. But each pair of distinct r-Sylow subgroups intersect trivially, so the total number of non-identity elements contained in all r-Sylow subgroups is

$$p^2(r-1).$$

Including the identity, this accounts for $p^2(r-1) + 1$ elements of G. Since $|G| = p^2r$, the number of elements not contained in any r-Sylow subgroup is

$$p^2r - \big(p^2(r-1) + 1\big) = p^2 - 1.$$

Each p-Sylow subgroup contains $p^2 - 1$ non-identity elements, and distinct p-Sylow subgroups intersect trivially. Hence there is room for at most one p-Sylow subgroup, contradicting the assumption that $n_p = r > 1$.

Case 3: p, q, r distinct: Assume that p, q, r are distinct primes and without loss of generality assume that $r < q < p$. As previously stated, if any of n_p, n_q, n_r are equal to 1, then there is a normal p, q, or r-Sylow subgroup and we are done. Thus suppose this is not the case, i.e. $n_p, n_q, n_r \neq 1$. Since p is the largest prime, n_p cannot be equal to q or r. Hence $n_p = qr$. Furthermore, since $q > r$, $n_q \neq r$ and hence $n_q \geq p$. And finally, $n_r \geq q$. Since distinct p-Sylow subgroups intersect trivially, the number of non-identity elements contained in all p-Sylow subgroups is

$$n_p(p-1) = qr(p-1).$$

Including the identity, this accounts for $qr(p-1)+1$ elements of G. Since $|G| = pqr$, the number of elements not contained in any p-Sylow subgroup is

$$pqr - \big(qr(p-1)+1\big) = qr - 1.$$

Each q-Sylow subgroup contains $q-1$ non-identity elements, and distinct q-Sylow subgroups intersect trivially. Hence the total number of non-identity elements contained in all q-Sylow subgroups is at least

$$n_q(q-1) \geq p(q-1).$$

But since $p > r$, we have

$$p(q-1) > r(q-1) = qr - r < qr - 1,$$

which is impossible since there are only $qr-1$ elements available outside the p-Sylow subgroups. This contradiction completes the proof.

$\square$

Returning to the proof of the exercise, by the Lemma we just proved, G is not simple. Thus, there exists a normal subgroup $N \lhd G$ with $N \neq \{e\}$ and $N \neq G$. Hence N either has prime order or has order of a product of two distinct primes. Similarly, the quotient group G/N has the same possible orders depending on the order of N. From the proof of Exercise 28, it follows that N and G/N are solvable, which implies that G is solvable.

$\square$

Exercise 0.23. (Exercise 30)

a) Prove that one of the Sylow subgroups of a group of order 40 is normal.

b) Prove that one of the Sylow subgroups of a group of order 12 is normal.

Proof. a) Let G be a group of order $40 = 2^3 \cdot 5$. Let $P = \mathrm{Syl}_2(G), Q = \mathrm{Syl}_3(G) \leqslant G$. Denote by n_2 and n_5 the number of $2-, 5-$Sylow subgroups of G. By Theorem 6.4,

$$n_2 \equiv 1 \mod 2,\ n_2 \mid 5 \quad \text{and} \quad n_5 \equiv 1 \mod 5,\ n_5 \mid 8.$$

Hence, $n_2 \in \{1+2k \colon k \in \mathbb{Z}_{\geq 0}\} \cap \{x \in \mathbb{Z}^+ \colon x \mid 5\} = \{1, 5\}$. From this alone, we cannot determine if G contains a normal $2-$Sylow subgroup or not. Thus, we see if G contains a normal $5-$Sylow subgroup. We have $n_5 \in \{1+5k \colon k \in \mathbb{Z}_{\geq 0}\} \cap \{x \in \mathbb{Z}^+ \colon x \mid 8\} = \{1\}$. Hence, $n_5 = 1$. Again, by Theorem 6.4, since all $5-$Sylow subgroups are conjugate and there is only one $5-$Sylow subgroup Q, we deduce that

$$gQg^{-1} = Q \quad \forall g \in G.$$

Hence, $Q \lhd G$.

b) Let G be a group of order $12 = 2^2 \cdot 3$. Let $P = \mathrm{Syl}_2(G), Q = \mathrm{Syl}_3(G) \leqslant G$. Then, by the using the same notation and logic of part (a),

$$n_2 \equiv 1 \mod 2,\ n_2 \mid 3 \quad \text{and} \quad n_3 \equiv 1 \mod 3,\ n_3 \mid 4.$$

Then $n_2 \in \{1+2k \colon k \in \mathbb{Z}_{\geq 0}\} \cap \{x \in \mathbb{Z}^+ \colon x \mid 3\} = \{1, 3\}$ and
$n_3 \in \{1+3k \colon k \in \mathbb{Z}_{\geq 0}\} \cap \{x \in \mathbb{Z}^+ \colon x \mid 4\} = \{1, 4\}$. If $n_3 = 1$, then we are done as we would

deduce that $Q \triangleleft G$. Thus, assume that $n_3 = 4$. Hence, there are 4 3−Sylow subgroups. These intersect trivially, so in total they contribute $4 \cdot (3-2) = 8$ non-identity elements to G. Observe that it is impossible to also have that $n_2 = 3$. If this was the case, then the 3 2−Sylow subgroups would contribute $3 \cdot (4-1) = 9$ non-identity elements to G which contradicts
$|G| = 12 \neq 8 + 9 + 1 = 18$. Therefore, we must have that $n_2 = 1$ and since all 2−Sylow subgroups are conjugate, and P is the only 2−Sylow subgroup, we have

$$gPg^{-1} = P \quad \forall g \in G.$$

Hence, $P \triangleleft G$.

□

Exercise 0.24. (Exercise 32) Let S_n be the permutation group on n elements. Determine the p-Sylow subgroups of S_3, S_4, S_5 for $p = 2$ and $p = 3$.

Proof. We utilize Thm 6.2 throughout the following derivations to justify existence. We will find the Sylow subgroups up to isomorphism. $\underline{(S_3;\text{ order } 3! = 6 = 2 \cdot 3)}$ Let $\mathbf{p} = \mathbf{2}$. The highest power of 2 dividing 6 is 2. Hence, let $P_2 \leqslant S_3$ be a 2-Sylow subgroups and hence $|P_2| = 2$. Thus, $P_2 \cong C_2$ (P_2 is the subgroup generated by any of the three transpositions of S_3). Let $\mathbf{p} = \mathbf{3}$. The highest power of 3 dividing 6 is 3. Hence, let $P_3 \leqslant S_3$ be a 3-Sylow subgroup, $|P_3| = 3$. Observe $(S_3 \colon P_3) = 2$, thus $P_3 \triangleleft S_3$ and the only nontrivial normal subgroup of S_3 is A_3. Hence, $P_3 = A_3 = \langle (123) \rangle \cong C_3$.
$\underline{S_4;\text{ order } 4! = 24 = 2^3 \cdot 3}$ Let $\mathbf{p} = \mathbf{2}$. A 2-Sylow subgroup P_2 has order $2^3 = 8$. Note that all elements of a 2-Sylow subgroup have an order of a power of 2. For example:

- Transpositions have order 2.
- The product of disjoint transpositions have order 2.
- 4-cycles have order 4.

Thus, there are no 3-cycles. We will construct a 2-Sylow subgroup. Start with disjoint the disjoint transpositions $(12), (34)$. These cycles generate the Klein 4-group,

$$V_4 = \{(1), (12), (34), (12)(34)\}$$

Thus far, we have order 4. Now consider the permutation $(13)(24)$. Conjugation by this element yields,

$$(13)(24)(12)(13)(24) = (43) = (34)$$

Hence, $(13)(24)$ swaps the two generators of V_4. Now the subgroup

$$P_2 = \langle (12), (34), (13)(24) \rangle$$

contains V_4 as a normal subgroup and an element acting by symmetry. Furthermore, $|P_2| = 8$. A group of order 8 containing a normal Klein 4-subgroup, plus an element that swaps its generators, is exactly the Dihedral group, D_4. Hence,

$$P_2 \cong D_4$$

Thus, all 2-Sylow subgroups of S_4 are conjugate and isomorphic to D_4. Let $\mathbf{p} = \mathbf{3}$. A 3-Sylow subgroup of S_4 has order 3 and thus is generated by any 3-cycle. Thus, $P_3 \cong C_3$. $\underline{S_5;\text{ order } 5! = 120 = 2^3 \cdot 3 \cdot 5}$ Since the 2-,3-Sylow subgroups of S_5 will have the same order as the 2-,3-Sylow subgroups of S_4, we have $P_2 \cong D_4$ and $P_3 \cong C_3$.

□

Exercise 0.25. (Exercise 35)
Show that there are exactly two non-isomorphic non-abelian groups of order 8. (One of them is given by generators σ, τ with the relations

$$\sigma^4 = 1, \quad \tau^2 = 1, \quad \tau\sigma\tau = \sigma^3.$$

The other is the quaternion group.)

Proof. Assume that G is a non-abelian group of order 8. Then there are no elements in G of order 8 since that would imply that G is cyclic and therefore abelian. Hence every non-identity element of G has order 2 or 4. If every element has order 2, then if $x, y \in G$, we have

$$\begin{aligned}
(\underbrace{xy}_{\in G})^2 &= e \\
xyxy &= e \\
xy &= y^{-1}x^{-1} \\
xy &= yx \quad \because y^{-1} = y,\, x^{-1} = x
\end{aligned}$$

But then G is abelian. Thus there exists at least one $g \in G$ of order 4. Hence, $(G\colon \langle g\rangle) = 2$ which implies that $\langle g\rangle \lhd G$. Then

$$hgh^{-1} \in \langle g\rangle$$

for all $h \in G$. Then since hgh^{-1} has order 4, we have $hgh^{-1} = g$ or g^{-1}. In the first case, $hg = gh$ which implies that G is abelian. Thus $hgh^{-1} = g^{-1}$. The quotient group $G/\langle g\rangle$ has order 2 and thus for every $h \in G/\langle g\rangle$, $h^2 = e \in G/\langle g\rangle \Rightarrow h^2 \in \langle g\rangle$. By construction of G, $|h| = 2$ or 4, so $|h^2| = 1$ or 2. Therefore,

$$h^2 \in \{e, g^2\}.$$

If $h^2 = e$, then $G = \langle g, h\rangle$ with $g^4 = e$, and $hgh^{-1} = g^{-1} = g^3$ which is precisely the definition of D_4. For the other case, $h^2 = g^2$, $g^4 = e$, and $hgh^{-1} = g^{-1}$ and this is precisely the quaternion group.

□

Exercise 0.26. (Exercise 36)
Let $\sigma = [123\cdots n]$ in S_n. Show that the conjugacy class of σ has $(n-1)!$-elements. Show that the centralizer of σ is the cyclic group generated by σ.

Proof. Recall the class equation is given by

$$(G\colon 1) = |G| = \sum_{x\in C}(G\colon G_x) = \sum_{x\in C}|\bar{x}|$$

with C a set of distinct representatives for each conjugacy class, G_x the isotropy group of x in G, and $\bar{x}$ denoting the conjugacy class of x. Hence, for this exercise, we consider $G = S_n$ acting on itself via conjugation and therefore $G_x = Z(x)$, the centralizer of x. Then we have

$$|S_n| = \sum_{\tau\in C}(S_n\colon Z(\tau)) = \sum_{\tau\in C}|\bar{\tau}|.$$

Therefore, since the sums are indexed by the same set C and are equal, we have

$$|\bar{\sigma}| = (S_n\colon Z(\sigma)) = \frac{|S_n|}{|Z(\sigma)|}$$

with $\sigma = (123\cdots n) \in S_n$ (we use parentheses to denote a permutation instead of brackets, as it is given in the exercise statement). it should be noted that, by definition, $|\sigma| = n$. Thus, if we can show that $Z(\sigma) = \langle\sigma\rangle$, our results will follow.
Let $\tau \in Z(\sigma)$. Thus $\tau\sigma = \sigma\tau$. Evaluating at 1,

$$(\tau\sigma)(1) = \tau(\sigma(1)) = \tau(2) = \sigma(\tau(1)).$$

Then, if we know what $\tau(1)$ is equal to, we can then compute $\tau(k)$ for any $k \in \{1, \ldots, n\}$. Suppose $\tau(1) = j$ for some $j \in \{1, \ldots, n\}$. Then,

$$\begin{aligned}
\tau(2) &= \sigma(\tau(1)) = \sigma(j) \\
\tau(3) &= \sigma(\tau(2)) = \sigma^2(j) \\
&\cdots \\
\tau(k) &= \sigma^{k-1}(j), \quad 1 \le k \le n
\end{aligned}$$

Furthermore, $\sigma^l(j) = j + l \mod n$. Hence,

$$\begin{aligned}
\tau(1) &= j \in \{1, \dots, n\} \\
\tau(2) &= \sigma(j) = j + 1 \mod n \\
&\cdots \\
\tau(n) &= \sigma^{n-1}(j) = j + (n-1) \mod n
\end{aligned}$$

For τ to be a permutation, these must be distinct. Thus, there are exactly n permutations that commute with σ. Then,

$$\begin{aligned}
\tau(1) &= \sigma^0 = \text{id}. \\
\tau(2) &= \sigma^1 \\
&\vdots \\
\tau(n) &= \sigma^{n-1}
\end{aligned}$$

are distinct. Hence

$$\begin{aligned}
Z(\sigma) &= \{\tau \in S_n \colon \tau\sigma = \sigma\tau\} \\
&= \{\sigma^k \colon 0 \leq k < n\} \\
&= \langle \sigma \rangle.
\end{aligned}$$

Therefore $Z(\sigma)$ is cyclic generated by σ and $|Z(\sigma)| = n = |\sigma|$. Then by the equality

$$|\bar{\sigma}| = \frac{|S_n|}{|Z(\sigma)|},$$

it follows that

$$|\bar{\sigma}| = \frac{n!}{n} = (n-1)!$$

Therefore, there are $(n-1)$-elements in the conjugacy class of σ.

□

Exercise 0.27. (Exercise 39)
Show that the action of the alternating group A_n on $\{1, \dots, n\}$ is $(n-2)$-transitive.

Proof. Let $J_n = \{1, \dots, n\}$. First, note that A_n acts transitively (1-transitive) on J_n since $A_n \cdot 1 = \{\gamma \cdot 1 \colon \gamma \in A_n\} = J_n$, i.e. only one orbit. Let $X_{n-2} = \{x_1, \dots, x_{n-2}\}, Y_{n-2} = \{y_1, \dots, y_{n-2}\}$ and set $A = J_n \setminus X_{n-2} = \{x_{n-1}, x_n\}$ and $B = J_n \setminus Y_{n-2} = \{y_{n-1}, y_n\}$. Since S_n is n-transitive, and thus $(n-2)$-transitive, there exists some $\gamma \in S_n$ such that $\gamma x_i = y_i$ for all $i = 1, \dots, n-2$. Furthermore, $\gamma(A) = B$. If γ is even (i.e. $\gamma \in A_n$), we are done. Thus, suppose $\gamma \notin A_n$. Let $(y_{n-1}\ \ y_n)$ be the transposition of the elements in B. Set

$$\sigma = (y_{n-1}\ \ y_n)\gamma.$$

Then σ is even since $(y_{n-1}\ \ y_n)$ and γ are odd. Thus,

$$\begin{aligned}
\sigma x_i &= (y_{n-1}\ \ y_n)\gamma x_i \\
&= (y_{n-1}\ \ y_n) y_i \\
&= y_i \quad \forall\, i = 1, \dots, n-2 \ \ (\because y_i \notin B = \{y_{n-1}, y_n\}).
\end{aligned}$$

Therefore, $\sigma x_i = y_i$ for all $i = 1, \dots, n-2$ and $\sigma \in A_n$. Hence, A_n acts $(n-2)$-transitively.

□

Abelian Groups

Exercise 0.28. (Exercise 42)
Viewing $\mathbb{Z}, \mathbb{Q}$ as additive groups, show that $\mathbb{Q}/\mathbb{Z}$ is a torsion group, which has one and only one subgroup of order n for each integer $n \geq 1$, and that this subgroup is cyclic.

Proof. We have

$$\mathbb{Q}/\mathbb{Z} = \{x + \mathbb{Z} \colon x \in \mathbb{Q}\}.$$

Let $x + \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ be an arbitrary nonzero element. Let $x = \frac{a}{b} \in \mathbb{Q}$. Then

$$b(x + \mathbb{Z}) = b\left(\frac{a}{b} + \mathbb{Z}\right) = a + \mathbb{Z} = 0 + \mathbb{Z} = 0.$$

Therefore, $x + \mathbb{Z}$ has finite order. Then, since $x + \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ was arbitrary, every element of $\mathbb{Q}/\mathbb{Z}$ is a torsion element and, therefore, $\mathbb{Q}/\mathbb{Z}$ is a torsion group. For each integer $n \geq 1$, define

$$H_n \left\langle \frac{1}{n} + \mathbb{Z} \right\rangle.$$

Then $n\left(\frac{1}{n} + \mathbb{Z}\right) = 1 + \mathbb{Z} = 0$. Thus, $\frac{1}{n} + \mathbb{Z} \in \mathbb{Q}/\mathbb{Z}$ has order dividing n. If $0 < k < n$, then

$$k\left(\frac{1}{n} + \mathbb{Z}\right) = \frac{k}{n} + \mathbb{Z}$$

and $\frac{k}{n} \notin \mathbb{Z}$. Hence, $\frac{1}{n} + \mathbb{Z}$ has order exactly n. Thus,

$$H_n = \left\langle \frac{1}{n} + \mathbb{Z} \right\rangle \leqslant \mathbb{Q}/\mathbb{Z}$$

is a cyclic subgroup of order n. Suppose $K_n \leqslant \mathbb{Q}/\mathbb{Z}$ is a subgroup of order n. Thus, let $x + \mathbb{Z}$ be a generator for K_n. Write $x = \frac{a}{b} \in \mathbb{Q}$ in lowest terms. Then $x + \mathbb{Z}$ has order $b = n$. Thus, every generator of a subgroup of order n has the form

$$\frac{a}{n} + \mathbb{Z}$$

with $(a, n) = 1$. Then a is invertible modulo n and thus there exists some $c \in \mathbb{Z}$ such that

$$ac \equiv 1 \mod n$$

Thus $ac = 1 + nk$ with $k \in \mathbb{Z}$. Hence,

$$c\left(\frac{a}{n} + \mathbb{Z}\right) = \frac{ca}{n} + \mathbb{Z} = \frac{1 + nk}{n} + \mathbb{Z} = \frac{1}{n} + \mathbb{Z}.$$

Therefore,

$$\left\langle \frac{a}{n} + \mathbb{Z} \right\rangle = \left\langle \frac{1}{n} + \mathbb{Z} \right\rangle = H_n$$

and thus $\mathbb{Q}/\mathbb{Z}$ has one subgroup of order n, namely H_n, which is cyclic.

□

Exercise 0.29. (Exercise 44)
Let $f \colon A \to A'$ be a homomorphism of abelian groups. Let B be a subgroup of A. Denote by A^f and A_f the image and kernel of f in A respectively, and similarly for B^f and B_f. Show that $(A \colon B) = (A^f \colon B^f)(A_f \colon B_f)$, in the sense that if two of these three indices are finite, so is the third, and the stated equality holds.

Proof. Consider the surjective homomorphism

$$\psi \colon A/B \to A^f/B^f$$

given by $a + B \mapsto f(a) + f(B) = f(a) + B^f$. This is well defined for if $a + B = a' + B$, then $a - a' \in B$. Thus, $f(a - a') = f(a) - f(a') \in B^f$. Hence, $f(a) + B^f = f(a') + B^f$. We show that $\ker \psi = A_f/B_f$. By definition,

$$\ker\psi = \left\{a + B \in A/B \colon f(a) \in B^f\right\}.$$

Let $a + B \in \ker\psi$. Thus,

$$\begin{aligned} f(a) \in B^f &\iff \exists b \in B \text{ such that } f(a) = f(b) \\ &\iff f(a-b) = 0 \\ &\iff a - b \in A_f. \end{aligned}$$

Since $a + B = (a - b) + B$, the coset of $a + B$ is represented by an element of A_f. Hence, $a + B \in A_f/B_f$. Thus, $\ker\psi \subseteq A_f/B_f$. Conversely, let $a + B_f \in A_f/B_f$. Thus,

$$a + B_f = x + B_f$$

for some $x \in A_f$. Then $a - x \in B_f \subseteq B$ and

$$f(a) = f(x) + f(a - x) = 0 + f(a - x) \in B^f.$$

Hence, $\psi(a + B_f) = B^f$ which implies that $f(a) \in B^f$, i.e. $a + B_f \in \ker\psi$. Therefore, $A_f/B_f \subseteq \ker\psi$ and $\ker\psi = A_f/B_f$. By the First Isomorphism Theorem,

$$\frac{A/B}{\ker\psi} = \frac{A/B}{A_f/B_f} \cong A^f/B^f.$$

Taking orders,

$$\begin{aligned} \left|\frac{A/B}{A_f/B_f}\right| = \frac{|A/B|}{|A_f/B_f|} &= |A^f/B^f| \\ \frac{(A \colon B)}{(A_f \colon B_f)} &= (A^f \colon B^f) \\ (A \colon B) &= (A^f \colon B^f)(A_f \colon B_f) \end{aligned}$$

as desired.

$\square$